

Bt503 By Tehreem&Arfa and Co workers

**this work is
inspired by
M.imran**

**wish you best of
luck for Grand
quiz**

**Admins
Tehreem&Arfa of
Msc zoology
nationwide**

Bt503

Lecture #1-3

Objective

By Arfa Amjad

MCQS

1. **Biotechnology** the application of science and technology to living organisms, as well as parts, products and models
2. **Environmental biotechnology** is the branch of biotechnology that addresses environmental problems, such as the removal of pollution, renewable energy generation or biomass production, by exploiting biological processes
3. The agricultural, industrial and environmental applications of biotechnology are **potentially** enormous.
4. Manufacturing process, waste management, pollution control three key points for **environmental biotechnology** interventions.
5. Environmental biotechnology deal with apparently **less dramatic** topic.
6. Resort to some form of alternation to bring about the desired **outcome**.
7. Enhance or optimize conditions for **existing biological** systems to make their activities happen faster or more efficiently.
8. **Massive** growth in global environmental market.
9. A prediction the market will have grown to **7400 billion** US dollars by 2025.
10. Greater general awareness, widespread adoption of sustainable best practice by industry, geo-political changes, and clean manufacturing applications are the **driving factors**.
11. Water treatment accounts for **25%** of the total global environmental market.
12. The **OECD** (2001) concluded that the industrial use of biotechnology commonly leads to increasingly environmentally harmonious processes and additionally results in lowered operating and/or capital costs.

Bt503

Lecture #4-7

Objective

By Arfa Amjad

MCQs

1. **Optimization** of the activities of particular organisms, or even whole biological communities.
2. The manipulation of **natural cycles** lies at the heart of much environmental biotechnology.
3. In general the use of biotechnology for environmental management relies on **mesophilic** microorganisms.
4. Some **microbial species** tolerate extreme conditions, like high salinity, pressures and temperatures.
5. **Extremophiles** could provide a way of developing alternative routes to many conventional chemicals or materials.
6. Thermophiles temperatures above **45°C**.
7. **Hyperthermophiles** temperatures above ~85°C.
8. In **PCR** DNA Polymerase of *Thermus aquaticus* (Taq Polymerase), *Pyrococcus furiosus* (Pfu Polymerase).

No 8

Cold environment are more common than **hot ones**

Average oceanic temperature is 1-3C

Vast areas are permanently frozen

Psychrophiles can tolerate severe circumstances

Halophiles survive in saline water

Number of halophile species accumulate KCL

Acidophiles thrive in condition of low PH, below 5

Extremozymes are capable of functioning below PH 1

Alkaliphiles flourish in environment at PH 9

Diverse degrative abilities are toluene degradation, cellulose degradation and chloromethane degradation

NO 9

Word xenos meaning foreign

degradation e

NO 9

Word xenos meaning foreign

Novel biopolymers are particular to environmental biotechnologists

Xenobiotics are compound that are man made

Catabolism is a xenobiot is degraded by enzymes not provide energy

Gratuitous degradation xenobiot resembles a natural compound

Soil microorganisms are versatiles

Operons are carried on **plasmids**

In bacteria genes coding are arranged in operons

NO 10

Chemicals resist **degradation**

4

Natural oestrogens are deactivated in humans by glucuronidation

Glucuronidation a conjunction of hormone with UDP glucuronate

UDP glucuronate make the compound more **polar**

Bacteria have B-glucuronidase enzyme to deactivate hormones

Glucuronidation is used to detoxify drugs

Oestrogen is heat labile and susceptible to VU exposure

Oestrogens effects are augmented by titanium dioxides

NO 11

5

Polychlorinated biphenyls (PCBS) are xenobiotics having high level of halogenation

A strain of PSEUDOMONAS PUTIDA known to degrade PCBS

Phthalates are substituted single ring phenols

Phthalates are used to manufacture polyester fibres, films and plastic bottles

Methanogenic Conortium of over 100 bacterial clones with capability to digest terephthalates

NO 12

NO 12

Movement of genes within and between organisms is **mobility**

Taking up of foreign DNA by bacteria is **transformation**

Extra chromosomal elements are **plasmids and viruses**

Genetic transfer through bacteriophages is **transduction**

Insertion may be into specific sites or random

Rearrangement of genetic material is stimulated by **transposons**

Evolutionary process occurs by deletions and insertions

Viruses which infect a wide host range **retroviruses**

Alfalfa mosaic virus and Ti plasmid of Agrobacterium tumefaciens are retroviruses

viruses **RNA** are more susceptible to mutations than DNA

A blurred distinction between **chromosomal and extra chromosomal elements**

NO 13

Gregor Mendel studied about genetic manipulation

In bacteria for genetic transfer are plasmids and **bacteriophages**

For eukaryotes their most plausible vectors are **eukaryotic viruses**

Artificial mutations of genome is by the introduction of novel gene

NO 14



Tolerance may be increased in culturing growth medium '

expansions **Catabolic** is to improve microbes ability to degrade

Speeding up the rate of mutation is done by **mutagens**

Bacterium tolerance is improved by standard microbiology

BT503 Topic: 15,16

1. **Genetic manipulation** is a very powerful and established technique of a gene into a specified organism by deliberate.
2. **Bacteria** and **Fungi** usually described as **recombinants**.
3. **Higher plants** and **animals** commonly described as **transgenic**.
4. Transfer of a gene from one organism to another is **foreign gene**.
5. The **recombinant** bacterium may also lose its new capability through normal transfer of genes between bacteria.
6. **Marker genes** are used for determining whether or not the transfer of DNA has been successful.
7. Isolated **DNA** is purified from **protein**.
8. Isolated DNA is precipitated out of aqueous solution by addition of **alcohol**.
9. On addition of the **alcohol** the DNA appears as a **white, semi-transparent material, coiling** out of solution. 
10. DNA may be collected by **centrifugation** and dried down.
11. **Restriction endonucleases** cut the DNA at specific sequence, either producing a **flush/blunt or staggered** end.
12. In **cloning** procedures other restriction nucleases cut the DNA at some distance from the recognize site.
13. Preparation of the **vector** is dictated by the type of end prepared for the insert DNA; **flush** or **sticky**.
14. The **foreign** DNA is incubated with the prepared vector in an aqueous solution containing various salts and **ligase** enzyme.
15. Fundamental requirements of all **cloning** procedures are **enzymes, solutions, equipment's, piece of DNA, cloning vector and recipient cell**.



BT503 Topic: 17, 18

1. **DNA** contains the coding sequence for a **gene**.
2. Sources of coding sequence for a gene; **genomic DNA, complementary DNA (cDNA) library, a product of PCR, chemically produced DNA**.
3. **Genomic libraries** are the collection of total Genomic DNA from an organism.
4. **Genomic** libraries are very useful to look for a specific **DNA** sequence.
5. The **non-coding** regions of eukaryotic genes are called **introns**.
6. **Prokaryotes** do not contain **introns**.
7. In eukaryotes, the first product of transcription from DNA is **heterogeneous nuclear RNA (hnRNA)**.
8. The non-coding regions or **introns** are removed by **mRNA**.
9. **PCR** is a technique which amplifies a piece of a piece of **DNA**.
10. The denaturation of double stranded DNA at elevated temperature **95 C**.
11. Annealing of the primer by **polymerase** usually at **72 C**.
12. **Thermus aquaticus** is a **thermophilic** microorganism from which polymerase has been isolated.

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BT503 Topic: 19, 20, 21

1. A cloning vector is a **plasmid** or a **bacteriophage (bacterial virus)**.
 2. **Selector marker** genes are indicators of genomic integrity and activity.
 3. A common design of a cloning vector carries **two** genes coding for **antibiotic** resistance.
 4. **Selector marker** genes normally contain a multi cloning site (MCS) to make **manipulation** easier.
 5. MCS is a cluster of sites for **restriction** enzymes.
 6. Viruses may be used as **vectors**.
 7. Transfer of naked viral recombinant vector to host cells is called **transfection**.
 8. Viral DNA vectors need phage particles to perform proper gene transfer is called **transduction**.
 9. **Transfections** have lower uptake rates as compared to transduction.
 10. **Expression vectors** have the required regulatory regions located before and after the foreign gene. **10**
 11. **Expression vectors** are sometimes a difficult, expensive or time consuming procedure.
 12. In expression vectors there are **reporter genes** to indicate whether or not the signals are switch on.
 13. In **expression cloning**, there are many circumstances which can be difficult to predict.
 14. **Antibiotic resistance gene, or b-galactosidase** are genes usually code for a trait or an enzyme.
 15. Reporter genes produce enzymes can cause the emission of light such as luciferase isolated from **fireflies**.
 16. **B-glucuronidase** is the reporter genes used in transgenic plants.
 17. Reporter genes can only be a guide to the process of **transcription** and **translation** process.
 18. The first step in screening is change in **antibiotic sensitivity** .
 19. The vector (plasmid) allows the insertion of foreign DNA for a **color test**.
 20. The second step in screening is usually to **probe** for a desired gene.
 21. The 3rd stage in screening is normally to analyze the **DNA** isolated from possible recombinants.
-

22. **Southern blot** is a procedure to scan cheaper and quicker a large number of samples.
23. In **southern blotting**, DNA fragments are resolved by **electrophoresis** on a gel.
24. The resolved fragments are then shifted to a membrane (**nylon or nitrocellulose**).
25. The nylon membrane is then **probed** by a piece of DNA complementary to the sequence of interest.
26. If the required sequence are present the band will show up on **autoradiography**.
27. Nothern blotting is used for RNA.
28. **Western blotting** is used for proteins.





BT503

Lec 22

- GE of **microorganisms** for use in environmental biotechnology.
- A strain of Escherichia coli into which was engineered some **15** genes originating from Pseudomonas.
- The **traditional method** requires the use of toxic chemicals with the associated safety measures and inherent pollution problem.
- Similar to recombinant technologies were investigated in early **1980's** by Amgen and Zeneca in UK.
- Yeast is **unicellular** eukaryote.
- Yeast, being unicellular eukaryote, has become popular for **both** (cloning and expressing eukaryotic genes.)
- Yeast cells are surrounded by a thick cell wall which must be removed to permit entry of **DNA** into the cell.
- Several types of **plasmid** vectors available for GE, some of which have been constructed to allow replication in both bacteria and Yeast.
- **Two** crossover events take place between homologous regions on vectors and yeast genome which effectively swap over a piece of host DNA with the plasmid DNA.

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BT503

Lec 23

- The insect virus, **Baculovirus**, has been shown to be the method of choice for the over expression of genes in many applications of molecular biology.
- Recombination in Baculovirus occurs by much the same way as in **yeast**.
- Replacement of **p10** proteins of Baculovirus by the gene for a scorpion neurotoxin, with a view to improving the insecticidal qualities of the virus.

BT503

Lec 24

- Many of the transgenic plants have been produced using the **Ti plasmid** transfer system of Agrobacterium tumefaciens and are often used together with the **35S CaMV** promoter.



- Genetically modified plants are called **transgenic plant**.
- GE in **agro biotechnology** is focusing on genetic modifications to improve crop plants with respect to quality, nutritional value and resistance to damage by pests and diseases.

BT503

Lec25 and 26

- Introduction into sugarcane of genes able to inactivate toxins produced by the bacterium, *Xanthomonas albilineans*, causing **leaf scald disease**.
- The method of **biolistic bombardment** may increase in popularity in line with improvements to plastid transformation.
- The most popular method of transforming plants is by the **Ti plasmid**.
- **Protoplast fusion** is a process whereby the plant cell wall is removed leaving the cell surrounded only by the much more fragile membrane
- The **DNA construct** contains regions of DNA which are complementary to the plant DNA to enable the inserted piece to recombine into the plant genome.
- Wild type plasmid contains genes which causes the transfer of a piece of DNA, **'T-DNA'**, into a plant cell, causing crown gall disease.
- The T-DNA' comprises genes which cause **crown gall disease**.
- The piece of T-DNA is flanked by **24 bp** direct repeats.
- Genes usually introduced more successfully on a second Ti plasmid during a co-infection with **Agrobacterium** carrying the plasmid containing the gene of interest.

BT503

Lec27

- The gene coding for the enzyme 5- enolpyruvylshikimate-3-phosphate synthase has been identified, isolated and inserted into a number of plants including **Petunias**



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- The piece of rDNA is flanked by **24 bp** direct repeats.
- Genes usually introduced more successfully on a second Ti plasmid during infection with **Agrobacterium** carrying the plasmid containing the gene of

2 / 3

BT503

Lec27

- The gene coding for the enzyme 5- enolpyruvylshikimate-3-phosphate synthase has been identified, isolated and inserted into a number of plants including **Petunias**

- A general strategy to protect plants from various agents has been proposed using **tobacco plants** as a model.
- The transgenic express the **iron-binding protein**, ferritin, in their cells which appears to afford them far ranging protection.
- 'Glyphosate' (**herbicide**) is an analogue of phosphoenol pyruvate and inhibits the enzyme 5- enolpyruvylshikimate-3-phosphate synthase.
- The gene was expressed behind a **CaMV promoter** and introduced using **A. tumefaciens**.
- **A. tumefaciens** has been to transfer the genes for mammalian cytochrome P450 monooxygenases, known to be involved in the **detoxification** of many xenobiotic including pesticides, into tobacco plants.

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BT503

Lec28

- Attack by **insects** not only causes damage to the plant but also provides a route for bacterial or fungal infection.
- To increasing resistance to sustained attack, the genes coding for the δ -endotoxin of *Bacillus thuringiensis* have been transferred **into plants**.
- Introduction by **A. tumefaciens** of the genes expressing antibodies to the coat protein of Tobacco Mosaic Virus (TMV).

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using **tobacco plants** as a model.

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- High expression and stability of the BT proteins whose genes have been introduced into **chloroplasts**.
- *Bacillus thuringiensis* tend to use codon richer in **thymidine and adenine** than the plant cell into which the gene is placed.
- Each Amino acid incorporated into a protein there is usually a choice of **three or four** codons all of which code for that same amino acid.

❖ **Noor-ul-Huda**

✓ **Bt503**

➤ **Chapter#29, 30,31.**

1. Bacteria communicate with each other by way of small diffusible molecules such as the

N_acylhomoserine lactose

2. May plant pathogens also cause_____.

Pathogenicity

3. Plants are susceptible to bacterial pathogens such as _____.

Erwinia carotovora

4. Enzymes are control by _____.

Acylhomoserine lactose

5. The rational behind using _____ for plants protection is to make transgenic plants.

Acylhomoserine lactose

6. _____ which colonies the surface of leaves.

Pseudomonas syringe

7. *Pseudomonas syringe* produce a protein which promotes the information of ice crystal just below _____.

0 ° C

9. _____ to simplify the genetic manipulation.

Escherichia coil

10. Salt tolerance in Tomatoes has been established by the introducing gene involved in _____.

Na⁺ /H⁺

11. Antiport occurred in _____.

Leaves

12. Gene guns the popular trees _____.

Liriodendron tulipifera

13. Initially the Gene was transferred to _____.

Arabidopsis thaliana

14. Chapters#32, 33,34,35

15. The fatty acid composition in the seeds has been modified to produce _____.

Triacylglycerol

16. Bioindicator are _____.

Living organisms

17. Plankton are rapidly changes taking place in the surrounding even the health of _____.

Aquatic flora

18. _____ utilized the moss.

Hasselbeck et al

19. Lichens are utilizing by the _____.

Quantitative estimation

20. _____ & _____ are frequently used to monitor air contamination.

Lichens, bryophytes

21. Increased level of _____.

SO₂,N₂

22. The diversity of species _____.

Phytoplankton

23. Zooplanktons are _____.

Alona guttata, cyclops

Lectures 36-42

Created by Tasmia

1. Variation in population of animals may indicate-----changes caused due to pollution into ecosystem. **Harmful**
2. Animal indicators also help in detecting the amount of----- (**toxins**) present in the tissues of animals.
3. Frog is also ----- (**indicators**)
4. Aquatic invertebrates tend to be----- (**bottom feeders**).
5. ----- (**Microorganism**) are often used as health indicators of aquatic and terrestrial ecosystem.
6. Microorganism when exposed to cadmium and benzene contaminants develop new proteins known as ----- (**stress protein**)
7. Microbial indicators can be used in a variety of way to detect the environmental pollution in water including the use of ----- (**bioluminescent**) bacteria.
8. In absence of metal pollution bacterium produce ----- (**blue pigmentation**)
9. (**Vogesella indigofera**)----- react to heavy metal.

known as ----- (**stress protein**)

7. Microbial indicators can be used in a variety of way to detect the environmental pollution in water including the use of ----- (**bioluminescent**) bacteria.
8. In absence of metal pollution bacterium produce ----- (**blue pigmentation**)
9. (**Vogesella indigofera**) ----- react to heavy metal.
10. Under the vicinity of hexavalent chromium the production of pigment is ----- (**blocked**)
11. Bioorganism are basically used to define the characteristic of a ----- (**biosphere**)
12. Biomonitors are used to get ----- (**quantitative**) information on the quality of environment.
13. Biomonitors are freely available all around the ----- (**world**)
14. Planktons are composed of organisms with ----- (**chlorophyll**)
15. Indian lentic ----- (**ecosystem**) were researched for planktons amid the mid twentieth century.
16. The occurrence of planktonic organism is identify with ----- (**resistance range**) in relation to abiotic ecological components.
17. Cynobacteria produce toxins which are harmful for ----- (**fish growth**)
18. Phytoplanktons are also known as ----- (**microalgae**).
19. Algae are quite ----- (**sensitive**) to contamination.
20. ----- (**Chlamydomonas reinhardtii**) was sensitive to cadmium, methylmercury and lead.

to abiotic ecological components.

17. Cynobacteria produce toxins which are harmful fo----- (fish growth)
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19. Algae are quite ----- (sensitive) to contamination.
20. ----- (Chlamydomonas reinhardtii) was sensitive to cadmium, methylmercury and lead.

19

21. Phytoplanktons are important source of pollutant transfer from water to upper trophic level and even ----- (humans).

LECTURE # 43. ZOOPLANKTONS

1. **Zooplanktons** are microscopic animals living near to the surface of the water body.
2. They are **poor swimmers**, instead relying on tides and currents as a transport mechanism.
3. They feed upon **phytoplankton, bacterioplanktons, or detritus** (i.e. marine snow).
4. Zooplanktons constitute a vital food source for **fish**.
5. They also play an important role as **Bio indicators** and help to evaluate the level of **water pollution**.
6. They are assumed to be a vital part in indicating **water quality, eutrophication**, and production of a **freshwater body**.
7. In order to determine the status of a freshwater body it is necessary to measure **seasonal variations** and **presence of zooplanktons**.
8. Zooplankton development and conveyance are subject to **abiotic** (e.g. temperature, saltiness, stratification, and pollutants) and **biotic** parameters (e.g. limitation of food, predation, and competition)

20**LECTURE # 44. EVIDENCES PERTAINING TO ZOOPLANKTONS**

9. Lake Mirik, in Darjeeling, Himalayas, on was polluted due to toxins let into the lake resulting in a **decreased pH** and an **increased acidity level**.
10. Brought on a **reduction** in the quantity of species and changes in species strength.
11. In this condition, **cladocerans** and **copepods** were found.
12. Cladocerans include **Bosmina, Moina, and Daphnia**.
13. Copepods include **Phyllodiaptomus and cylops**.
14. Trichopria tetrad could be utilized as **contamination indicators** as they have been seen in lakes rich in **phosphorus** and other heavy metals. This species was obtained in the past in sewage-contaminated tanks.
15. Variation in the population of copepods, seasonally in various water bodies; density was **highest** in the rainy season, while it **reduced** in summers due to high temperatures.
16. Though Zooplankton may be present in an **extensive** variety of ecological conditions, yet disintegrated oxygen, temperature, salinity, pH, and other physicochemical parameters are **restricting** elements.

LECTURE # 45. AQUATIC INSECTS AS ENVIRONMENTAL INDICATORS

17. **10** orders of insects inhabit freshwater system
18. odonta commonly known as **dragon fly**
19. Coleoptera (**beetles**)
20. Ephemeroptera (**May flies**)
21. Plecoptera (**Stoneflies**)
22. Trichoptera (**Caddis flies**)
23. Megaloptera (**Dobson flies**)
24. Lepidoptera (**Moths & Butterflies**)
25. Neuroptera (**Lacewings & Mantid flies**)
26. Diptera (**Common flies**)
27. Hemiptera (**True bugs**)
28. Insects are such an integral part of the ecosystem that scientists are using them to gauge the health of rivers, this process is called **bio assessment**.
29. Bio assessment is also known as **bio monitoring**.
30. The process of monitoring the animals in an ecosystem to gauge the health of the environment they are in a specific type of bio monitoring that uses specific insects is called **EPT**.
31. EPT stands for **Ephemeroptera, Plecoptera and Trichoptera** which are three groups of insects. They are aquatic insects.

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LECTURE # 46. POLLUTION AND POLLUTION CONTROL

32. **Pollution**, the most frequently talked about of all environmental Problems, yet in many respects, it can often remain one of the least understood.
33. The diverse nature of potentially **polluting substances** can lead to some confusion..
34. “ ‘Pollution of the environment’ means pollution of the environment due to the **release** (into any environmental medium) from any process of substances which are capable of causing harm to man or any other living organisms supported by the environment.”
35. The **escape** of any substance capable of causing harm to man or any other living organism supported by the environment.”

LECTURE # 47. CLASSIFYING POLLUTION

36. The **diverse** nature of potential pollutants makes their systematization difficult in absolute terms.
37. Classification may, for example be made on the basis of **(All of the given below)**
- The chemical nature of the substance,
 - Physical nature of the substance,
 - Its source,
 - The environmental pathway used,
 - The target organism affected, or
 - Simply its gross effect.
38. The consideration of a pollutant's properties is a particularly valuable approach when examining real-life pollution effects. E.g., • toxicity; • persistence; • mobility; • ease of control; • bioaccumulation; • chemistry. **(All of the above)**

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LECTURE # 48. TOXICITY

39. **Toxicity** represents the potential damage to life and can be both short and long term.
40. It is related to type of the **pollutant**, **concentration**, and the **time of exposure**.
41. Intrinsically **highly toxic** substances can kill in a **short** time, while **less toxic** ones require a **longer** period of exposure to do damage.
42. In the case of low concentration exposure, some pollutants may also have an effect on an organism's **behavior** or its susceptibility to environmental stress over its lifetime.
43. **Availability** also features as an important influence, both in a gross, physical sense and also in terms of its biological availability to the individual organism, together with issues of its age and general state of health.

LECTURE # 49. PERSISTENCE & MOBILITY

- 44. **Environmental persistence** is a particularly important factor in pollution and is often linked to mobility and bioaccumulation.
 - 45. Highly toxic chemicals which are environmentally unstable and breakdown rapidly are **less harmful** than persistent substances, even though these may be intrinsically less toxic.
-

23

- 46. **Mobility:** The tendency of a pollutant to disperse or dilute is a very important factor in its overall effect, since this affects **concentration**, as well as **exposure** to the living organisms.
- 47. Some pollutants are not readily **mobile** and tend to remain in 'hot-spots' near to their point of origin. Others spread readily and can cause widespread contamination.
- 48. Whether the pollution is **continuous** or a **single event**, and if it has arisen from a single point or multiple sources, form important considerations.

Aish

Lesson no 50-55

- ❖ Post pollution survival of a given environment depends on the maintenance of its natural cycle.
- ❖ Pollution effect are not always entirely defined by the initial nature of the contamination.
- ❖ Geology of the site may also influence the out-come.
- ❖ The attenuation of pollutants by the permitting to become physically-spread out.
- ❖ Air environment gives good dispersal and dilution of gaseous emission.
- ❖ Lighter particles spread easily.
- ❖ Good dispersal and dilution potential in large bodies of water.
- ❖ Movement through the soil represent another opportunity for the dilute and disperse approach.

Lesson no 50-55

- ❖ In some cases containment may be solution form.
- ❖ Some pollutants when present in very small amounts with in the enivornment, can be taken up by living organisms and become concentrated in the tissues over time.
- ❖ Bioaccumulation refers to how pollutants enter a food-chain.
- ❖ Biomagnification refers to the tendency of pollutants as the move from on trophic level to the next.
- ❖ There is sometimes a tendency for contamination to be considered somewhat in isolation from its context.
- ❖ The nature of the soil or water which harbours the pollution can have a major effect on the actual expressed end-result.

BI503: Environmental Biotechnology

Lecture # 57 - 63.

* Lecture # 57.

1. The toxicity of metal is related to their place in Periodic table.

2. Toxicity of metals reflect their affinity for amino acid Sulphate group.

3. Group IA reactivity is $\text{Na} < \text{K} < \text{Rb and Cs}$.

4. Group IB reactivity is $\text{Cu} < \text{Ag} < \text{Au}$.

5. Group IIA reactivity is $\text{Mg} < \text{Ca} < \text{Sr} < \text{Ba}$.

6. Group IIB reactivity is $\text{Zn} < \text{Cd} < \text{Hg}$.

7. Group IIIA reactivity is $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$.

8. Type A metals are Less toxic than type B.

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* Lecture # 58.

1. Karachi is the industrial background of Country.

2. Water is present on our planet 71%.

3. 98% water is sea or salty water.

4. 2% water is Fresh water.

5. we use less than 1% of fresh water in daily use.

6. Leyari river is the only river which flows in Karachi city.

7. Korangi domestic area is the best place to remove harmful chemicals.

8. Textile and pharma industries throw waste in mahtise river.

9. 1000 of fishes are dying due to polluted water.

* Lecture # 59.

1. Bioremediation is subjected by bacteria

2. Yeast produce arial lypoa which may be able to metabolize metal directly from

3. Removal of H₂S by biological means was first discussed in 1920.

4. For air contamination Temperature is required 15-30 °C

5. For air contamination PH is required 6-9

* Lecture # 60.

1. Biopilters are relative large container made of Cast Concrete, metal or durable plastic

2. In biopfilter gas to be treated to forced or drawn through the Filter

3. Biopfilter can reduce odor release by 95% or more

* Lecture # 61.

1. Biotracking filters represents the intermidate technology between biopfilters

and bioscrubbers.

2. Biotracking filter medium has inert material
Slag or clinker

* Lecture # 62 :-

1. The Bio Scrubber is not itself truly a biological treatment system

2. Bio Scrubbers are most appropriate for hydrophilic compounds like acetone or methanol

3. In bioscrubber the actual process of biodegradation take place in Secondary bio reactor

* Lecture # 63.

1. Treatment of gases other than biotechnology are Adsorption, Absorption, ~~incineration~~ incineration and ozonation etc

Lecture 65, 65, 66,67

- Absorbing the compound in a suitable liquid; this may **oxidize** or neutralize it in the process.
- Activated carbon preferentially **adsorbs** organic molecule.
- **High temperature** oxidation effective against most contaminants, but costly.
- Use of ozone to oxidize some contaminants, like **hydrogen sulphide**; effective but can be costly.
- no **carbon monoxide** production is the Main advantages of biotechnological approaches to the issue of air contamination.
- Much of the focus of **environmental** biotechnology centers on the **remediation** of pollution or the treatment of **waste** products.
- this tends to form the **natural** constituency of the science.
- safe processes with highly 'green' **profile**, robust and **tolerant** of fluctuation. are the Main advantages of biotechnological approaches to the issue of air contamination.
- **Clean technology** is the mechanism by which pollution or waste may be reduced at source are varied.
- **Biotechnological** interventions are principally limited to the technology or processes, though they may also prove **instrumental** in permitting **procedural** change.
- The main areas in which biological means may be relevant fall into **three** broad categories.
- The field is a **fast evolving** one and many more types of biotechnological interventions are likely in the future, especially where commercial **pressures** derive a competitive **advantage**.
- Replacement of **existing** chemical methods of production with those based on **microbial** or enzyme action is an important potential area of **primary pollution** prevention.
- One role in which the use of **genetically** modified organisms could give rise to significant environmental **benefit**.
- Biological synthesis, either by whole organism or by **isolated enzymes**, tends to operate at a **lower** temperature.
- **high** enzymatic specificity, gives a much purer **yield** with fewer **by-products**.
- In the cosmetics sector, high demand for **isopropyl** myristate used in moisturizing creams.
- The **conventional** method for its manufacture runs at **high** temperature and pressure.
- An alternative approach, enzyme-based esterification.



Lecture 68

- **Amylase** enzymes from **malt** extract to degrade starch-based sizes for cheap and effective reduction of fabric stiffness and **improvement** to its drape.
- novel enzymatic methods provide a **fast** and **inexpensive** alternative to traditional flax extraction by breaking down.
- the **woody** material in flax straw, **reducing** the process time from **7 to 10** days, down to a matter of hours.
- The **enzyme-based** retting processes available for use on hemp.
- flax produce finer, cleaner **fibres**.
- biodegradable **polymeric** fibers which can be synthesized using modified soil **bacteria**, avoiding the current **persistence** of these materials in **landfills**.
- **Bio-scouring** for wool and cotton uses enzymes to remove dirt, wax, fats, etc.
- **Bio-bleaching** uses **enzymes** to fade materials, avoiding both the use of caustic agents and the **concomitant** effluent treatment problems. **30**
- Biological **catalysts** have also proved effective in shrink-proofing wool, improving quality while **ameliorating** the wastewater produced and **reducing** its treatment costs.
- **Bio-polishing** involves enzymes in shearing off cotton microfibers to improve the material's **softness**.
- **Bio-stoning** has been widely adopted to produce 'stone-washed' **denim**, with enzymes being used to fade the fabric.
- original **pumice** stone method, which had a higher water **consumption**.
- Incorporation of **absorbers** and microbes within a **geotextile** produced for use in **land** management around railways.
- Soaking up and subsequently biodegrading diesel and **grease**, the textile directly reduces **ground pollution**, provides safer working conditions and **reduces** risk of **fire**.



Lecture 69

- The leather industry has a lengthy history of using enzymes.
- In the bating process, residual hair and epidermis, together with non-structural proteins and carbohydrates, are removed from the skins.
- Traditionally, pancreatic enzymes were employed.
- Around 60% of the input raw materials in leather manufacturing ultimately ends up being discarded.
- enzyme additions have long been used to help manage this leather waste.
- Upsurge in the use of microbially-derived biological catalysts, cheaper and easier to produce.
- Chemical methods for unhairing hides dissolve the hairs.
- Combining chemical and biological catalysts significantly lessens the process time and quantities of reagents used.
- The enzymes also make intact hair recovery a possibility, opening up the prospect of additional income from a current waste. **31**
- It has been estimated that, in the UK, for a yearly throughput of 400 000 hides.
- enzymatic unhairing offers a reduction of around 2% of the total annual running costs (BioWise, 2001).
- Conventional treatments produce both airborne VOCs and surfactants.
- The use of enzymes not only gives better results, consistent quality, better final color and superior dye uptake, but also considerably reduces VOC and surfactant levels.



Bt503

Lecture #71-77

Objective

By Shazine Noor

MCQS

1. **Disastrous** outcome of Australia's attempts to use the Cane Toad (*Bufo marinus*) to control the **cane beetle**.
2. Essence of the specifically environmental contribution of this type of **bio intervention** lies in its ability to obviate the need for the use of polluting chemicals.
3. **Biotechnology** per se is not a central, or even necessary, requirement for all of biological control, as many methods rely on whole organism predators.
4. **Bio-controls** are often highly target specific, reducing the danger to other non-pest species.
5. A large number of insects pose a threat to crops & thus represent an economic concern (**global insecticide market estimated ~\$10 billion (US) / year**).
6. Success is much more dependent on a thorough understanding of the **life-cycles** of the organisms.
7. **Classical biological control** requires the importation of natural predators and is principally of use when the pest in question is newly arrived in an area.
8. Third method, augmentation, refers to means designed to bring about the increase in effectiveness of **natural enemies** to a given pest.
9. Simply of **artificially rearing** them in large numbers or may extend to more **intensive** and **sophisticated** measures like the modification, either by selective breeding or genetic manipulation, of the predator.
10. **Augmentation** is, obviously, a highly interventionist approach and relies on a regime of continual management to ensure its effectiveness.
11. Augmentation of **entomopathogenic** parasitic nematodes.

12. A role for the engineered application of biologically derived chemicals in this sector. E.g., **Azadirachta indica**, the neem.
13. The compound **azadirachtin** has been identified and isolated from the plant and it has been shown to have broad spectrum insecticidal properties.
14. **Semiochemicals** are natural messenger substances which influence growth, development or behavior in numerous plant and animal species E.g., pheromones.
15. **Crops worldwide** suffer severe damage as a result of a number of **pentatomid insects**, amongst which are several of the common brown bugs of North America (**Euschistus spp**).
16. A major part of **bio-control** involves thorough understanding of their migration patterns, a pheromone **methyl 2E,4Zdecadienoate**.
17. A blend of **Z5,E7-dodecadienal** and **Z5,E7-dodecadienol**, be deployed at US ports of entry.
18. In the case of another pentatomid, **Nezara viridula**, the use of this approach to biological control failed.
19. The **Siberian moth** **Dendrolimus superans** is a **vigorous defoliating pest** of northern Asian **coniferous forests**.
20. Genetic engineering of its **gut symbionts** to produce a reduced tolerance of environmental stress.
21. The **bio-substitution** of many of today's polluting substances or materials is a major potential avenue for the **environmental biotechnology**.
22. The **pollution** of many inland and coastal waters around the world is a well appreciated environmental problem and wider use of **non-toxic**.
23. Readily **biodegradable alternative products** could make a huge difference. In case of bio-lubricants, cost is a major issue.
24. **Inevitably**, users need to be convinced of the deliverable commercial benefits, the potential market is **enormous**.

- 25.The **petrochemical industry** has sought to meet the growing demand for more environmentally friendly products by developing biodegradable lubricants based on crude oil.
- 26.**No denying** the **burgeoning** interest in bio-lubricants, the actual machinery to be lubricated is extremely expensive.
- 27.A few equipment operators are willing to risk trying these new, substitute oils, they are **seldom willing** to guarantee their performance.



34

Bt: 503

Samad Zahara

Topic: 78

M.C.Q.S

1. Biomimetics is a field in which principle from engineering, chemistry and biology are applied to the synthesis of materials, synthetic systems or machines that have function.
2. According to US Naval sources, microfouling by surface-adhering biofilms can increase drag on warship upto 20% **35**
3. Modeling of shark skin and the subsequent development of a novel biomimetics coating technology known as Sharklet
4. The presence of microfouling by ~~surface~~ barnacles and other larger organisms can add 60%.
5. Zoospores settlement rate less than 85% that of conventional surface.

h, Bt: 503 Topic:

Topic:- 79:-

6. The use of eco-building materials
formed from hemp, hay, straw.

7. Thermafleece: a high density
product manufactured from
British Sheep wool.

8. Walls made from eco-material
have been found to a particularly
effective at sound suppression.

9. Conventional construction and
demolition waste possess
a considerable disposal
problem. **36**

← → Topic:- 80

10. Primary treatment the removal of
solids only

11. The removal of settleable solids
and part of organic matter Sec
donary treatment.

12. Removal Efficiency = E

13. Effluent Concentration of the
Pollutants = C_e.

15. 14 Removal of Specific Pollutants that were not removed in the Secondary treatment is Tertiary treatment.

Topic:- Q1

15. How many types of the WWT operations, Process & System

Physical unit process Chemical unit

Process, Biological unit Process.

37

16. Treatment methods in which removal or conversion of the Contaminants occurs by the addition of chemical products due to chemical reaction known as chemical unit Process.

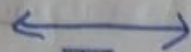
Topic:- Q2:-

17. What type of the bacteria act the bottom of the Pond anaerobic bacteria.

18. BOD in Suspended Particulate.

19. Phosphoreous Precipitation occurs in condition of high PH or through the addition of metabolic salts called Precipitation.

20. Conversion of BOD suspension into soluble BOD by means of enzymes, allowing its stabilization called hydrolysis.



Topic: 83, 84

38

21. The Oxygen required by aerobic bacteria is supplied by algae through Photosynthesis. This is known as Facultative Ponds.

22. 50 to 65% of the BOD is converted into anaerobic Pond.

23. In Maturation Ponds the main Object is removal of Pathogenic Organisms.

24. Photosynthetic activity is high leading to high

Completely mixing is done in Liquid Medium.

Concentration due to high pH.

Topic: 85

25. Waste water is applied in Shallow basins.

39

26. Free-Settled is applied below the solid surface.

27. Aquatic-based systems composed of shallow basin or channels in which aquatic plants grow.

Topic: 86

28)

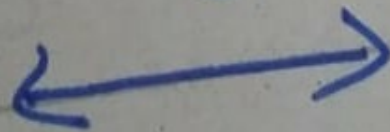
BOD is converted anaerobically by bacteria dispersed in the reactor.

40

29.

The nitrates found in the aerobic zone are used in the Respiration in the anoxic zone facultative microorganisms.

30.)



BT 503

Kafiza Fariha Shah

Lec#86

- 1: UASB is abbreviation of _____. (**Up flow anaerobic sludge blanket reactor**)
- 2: BOD is converted anaerobically by _____ in the reactor. (**bacteria dispersed**)
- 3: The upper part of the UASB reactor is divided into _____ (**settling and gas collection zones**).
- 4: The settling zone allows the exit of the _____. Amongst the gases formed is _____. (**clarified effluent, methane**)

Lec#87

- 5: The biological stage comprises two units _____. (**aeration tank (reactor) and secondary sedimentation tank**).
- 6: Septic tanks also a form of _____ treatment. (**primary**)
- 7: The biological reactor incorporates an _____, either upstream or downstream of the aerated zone. (**anoxic zone**)

41

8: The _____ formed in the aerobic zone are used in the respiration of facultative _____ in the anoxic zone. **(nitrates, microorganisms)**

Lec#88

9: Low availability of substrate (BOD) for the bacteria, which makes them undergo _____ and leave the system stabilised. **(self-digestion)**

10: Similar to the previous system but with the difference that a higher _____.
(BOD load is applied).

42

11: The effluent from the secondary sedimentation tank is _____ to the filter in order to _____. **(recirculated, dilute the influent)**

12: Besides being a support medium for biomass growth, the granular material acts also as _____. **(a filter medium).**

13: _____ are necessary to eliminate the excess biomass accumulated. **(Periodic washings)**

14: The biomass grows adhered to a support medium, which is usually composed by a _____ **(series of discs)**.

15: The discs, partially immersed in the _____, rotate, exposing their surface alternately to _____. **(liquid, liquid and air)**

Lec#89

16: Mainly intended for the removal of _____. **(Coarse solids, Gri)**

17: The removal of _____ is frequently done by screens or racks. **(coarse solids)**

18: In the screening, material with dimensions larger than the spaces between the bars is removed. There are _____ **(coarse, medium, and fine screens)**.

19: The removal of sand contained in the sewage is done through _____. **(grit chambers)**.

20: The sand removal mechanism is simply by _____. Organic matter stays in suspension and goes on to the downstream units. **(Sedimentation)**

21: The basic purpose of grit removal is to eliminate or reduce the possibility of obstructions in _____ **(piping, tanks, orifices, siphons, etc)**

Lec#90

22: Primary treatment aims at the removal of: _____ (**• settleable suspended solids • floating solids**).

23: The sedimentation tanks can be _____ (**circular or rectangular**).

24: The mass of solids accumulated in the bottom is called _____. (**raw primary sludge**)

25: The efficiency of primary treatment may be enhanced by the addition of _____ (**coagulants**).

26: Coagulants may be _____ or other, aided or not by a polymer. (**aluminium sulphate, ferric chloride**)

44

27: Phosphorus may be also removed by _____ (**precipitation**).

28: The septic tanks and their variants, such as Imhoff tanks, are basically _____.
(sedimentation tanks)

Lec#91

29: The secondary treatment processes are conceived in such a way as to accelerate the _____ that naturally occur in the receiving bodies. **(decomposition mechanisms)**

30: A great variety of microorganisms take part in the process: _____.
(bacteria, protozoa, fungi and others).

31: The microorganisms convert the organic matter into _____. **(carbon dioxide, water and cellular material)**

45

Lec#92

32: When facultative ponds receive raw sewage, they are also called _____.
(primary ponds).

33: The process of facultative ponds is the _____ only on natural phenomenon.
(simplest, relying)

34: The presence of _____ is necessary in aerobic respiration, and it is supplied to the medium by the photosynthesis carried out by _____. **(oxygen, the algae)**

35: Bacteria → _____ **(Respiration)**

36: Photosynthesis is higher near the _____ **(water surface)**.

37: The process of photosynthesis is natural, so does not require _____. Slow rate of organic matter stabilization, so need more time. **(any equipment, more than 20 days)**

46

38: Effective photosynthesis needs a _____, so area required is one of the largest.
(large exposure surface area)

❖ Smart doll/Andila➤ BT 503❖ Chapter#93, 94,95,96.

- 1) _____ require a large area.
Facultative ponds
- 2) Is the system of _____ ponds.
Anaerobic
- 3) Facultative ponds is also _____.
Secondary ponds
- 4) _____ practically does not occur.
Photosynthesis
- 5) Anaerobic bacteria have a _____ & _____.
Slow metabolic, reproduction
- 6) _____ there is only one partial decomposition of organic matter.
2-5
- 7) _____ BOD removal represents a large contribution.
50-70%
- 8) Facultative ponds receives a load of only _____ of the raw sewage load.
30-50%
- 9) Land savings in the order of _____ are obtained.
1/3
- 10) In Facultative aerated lagoons the oxygen is supplied by mechanical equipment called _____.
Aerators
- 11) Aerators having a _____.
Vertical axis
- 12) A large concentration of bacteria in the liquid medium, leads to large organic matter- _____ contact.
Biomass
- 13) Sedimentation pond have short detention times, around _____.
2 days
- 14) The ponds are shallow _____.
Less than 1.0m depth.
- 15) The photosynthesis activity is _____.
High
- 16) _____ removal also occurs as a result of high PH.
Phosphorus

❖ Chapter#97, 98,99,100

- 17) The main objective of maturation pond is the removal of _____.
Pathogenic organisms
- 18) The maturation pond is designed such as _____.
Temperature, PH, food storage etc.
- 19) Coliform removal efficiencies _____.
Up to 99.99%
- 20) Physical _____.
Settling, filtration, dehydration
- 21) Chemical _____.
Oxidation and chemical reactions, precipitation, adsorption
- 22) Biological _____.
Bio degradation and predation
- 23) The Phosphorus of wastewater can be done by method _____.
Sprinklers, furrows, graded border
- 24) Wastewater applied is not controlled by _____.
Crop
- 25) It is a natural system with the _____.
Highest efficiency
- 26) Vegetation growth _____ occur.
May or may not

BT503 lesson 101 to 107 mcqs

1. In this system **pre settled** and **pre treated sewage** is applied below ground level
2. The infiltration sites are prepared in **buried excavation.**
3. The sewage precoltes through **unsaturated soil**
4. Infiltration trenches or **pits**
5. The system are normally use **septic tanks**
6. Watse water flows genetaly downward where **biochemical** reaction takes place
- 7.. The effulent is collected at lower end by **drainage channels**
8. The soil should have **low permeability**
9. The slope should be moderate **2-8%**
10. The use of **vegetation** is essential for increasing absorption
11. Surface flow **wetland** natural appearance
12. Emergent **Typha**
13. Floating **eichornia**
14. Submerged **elodea**
15. Water depth is between **0.6-0.9m**
16. Water zones **1.2-1.5m**
17. Subsurface flow wetland are also called **vegetation submerged**
18. Medium height **0.5-0.6m**

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11. Surface flow **wetland** natural appearance
12. Emergent **Typha**
13. Floating **eichornia**
14. Submerged **elodea**
15. Water depth is between **0.6-0.9m**
16. Water zones **1.2-1.5m**
17. Subsurface flow wetland are also called **vegetation submerged**
18. Medium height **0.5-0.6m**
19. Water depth **0.4 - 0.5m**
20. Large parts of subsurface is **anaerobic**
21. The most classical conception of constructed **horizontal flow**
22. Widely used in rural areas is **reactors septic tanks**
23. In the imhoff tanks setting occur in **upper compartment**
24. Liquid flow is **upward**
25. BOD load applied per unit volume is **very high**
26. in **UASB** reactors biomass dispersed in liquid
27. Three phase separators **solid liquid gas**
28. Gas is collected in **upper part**
29. Hydraulics detention time **6 to 10 hour**
30. The **sludge** production is very low
31. BOD removal efficiently is around **70%**

- The sludge production is **very low** in **UASB**.
- To reach desired efficiency, some form of post-treatment must follow the **UASB reactors**.
- BOD removal efficiency is around **70%**
- Activated sludge systems may be classified...

According to the sludge age:

Conventional activated sludge

Extended aeration

- **ACCORDING TO FLOW SLUDGE SYSTEM MAY BE CLASSIFIED**

___ CONTINUOUS FLOW & INTERMITTENT FLOW

- **TREATMENT OF SLUDGE SYSTEM -- REMOVAL CARBON, REMOVAL OF CARBON And nutrients.**

50

- If excess sludge not stabilize aerobic stage it pumped back to the **___ anaerobic stage.**
- If the storage of bacteria could be returned to the aeration unit, the concentration of bacteria will be **increased**.
- Storage of bacteria increase the concentration of bacteria is basic principle of **___ activated sludge**.

basic principle of ____ activated sludge .

- ____ are essential items for the activated sludge system
.....(aeration tank, settling tank, pump , removal of biological sludge) .
- The biomes can be separated in ____ secondary sedimentation tank .
- The bioms stay in the system for longer time ____ due to sludge recirculation

- The retention time of the solids in the system is called **sludge age**
- Solid retention time is **4-10 days** **52**
- The biological excess sludge can be wasted directly from the **reactor.**

~~Topic~~

Topic 112 and 113

- ① High Biomass containing sludge has long Retention Period of 4-10 days
- ② If retention period is extended to 18-30 days it is called Extended aeration
- ③ In Extended aeration cellular organic matter is converted into carbondioxid and water through Respiration
- ④ Intermittent^{flow} of the liquid/sewage and operation also called a Sequencing Batch Reactor (SBR)
- ⑤ In extended aeration mode, The tank also incorporates the role of the Sludge digestion Unit
- ⑥ Normal treatment cycle in intermittent operation consists of following stages
* Fill * React * Settle, * Draw * Idle

53

Objective

By Tehreem Fatima

MCQS

1. The **activated sludge system** is capable of producing a satisfactory conversion of ammonia to nitrate (**nitrification**).
2. Only **ammonia** and not nitrogen is removed.
3. **Biological nitrogen removal** is achieved in the absence of **dissolved oxygen**, but **presence of nitrates**.
4. A group of bacteria uses the **nitrates** in their respiration process, converting them to **nitrogen gas**.
5. In activated sludge systems where **nitrification occurs**, it is interesting to induce **denitrification**.
6. Savings in oxygen (**energy economy in the aeration**), as under anoxic conditions, **facultative anaerobic bacteria** remove BOD by using the nitrate in their respiratory processes.
7. Savings in **alkalinity** (preservation of the buffering capacity).
8. During nitrification, **H⁺ ions** are generated which can lead to a decrease in the **pH**.
9. **Denitrification consumes H⁺** and generates alkalinity, partially compensating the pH reduction.
10. Operation of the secondary **sedimentation tank** (to avoid rising sludge).
11. Nutrient control (**eutrophication**).
12. The reduction of the nitrogen levels is important when the effluent is discharged into sensitive water bodies that are subjected to **eutrophication**.
13. The **anaerobic zone** gives good conditions for the development or selection of a large population of **phosphorus accumulating organisms**.
14. When the biological excess sludge is wasted from the system, **phosphorus is removed**.

15. Higher efficiencies of phosphorus removal.

Addition of coagulants (metallic ions): phosphorus precipitation

Effluent filtration: removal of the phosphorus present in the suspended solids in the effluent

Combination of the addition of coagulants and filtration.

55

III

BT-503 - Read-only



Read Only - You can't save changes to t...



BT 503

Fariha Shah

Lec#116

- 1: Aerobic units are____, in which the biomass grows attached to a support medium. **(biofilm reactors)**
- 2: _____consists of a coarse material bed, such as stones, gravel, etc, over which the wastewater is applied. **(Low rate trickling filter)**
- 3: With the passage of the wastewater, there is a contact between the _____. **(microorganisms and the organic matter)**
- 4: Wastewater is usually applied through rotating____. **(distributor)**
- 5: the primary function of the filter is_____ **(not to filter).**
- 6: The diameters of the stones are of the order of a few_____ **(centimeters)**
- 7: With increase in biofilm mass, _____, increasing the liquid velocity through the pores. **(the empty spaces decrease)**
- 8: This is a natural form of controlling the _____in the medium. The dislodged sludge must be removed by the. **(microbial population, secondary sedimentation tank)**
- 9: The applied BOD load per unit area and volume is____, therefore, food availability is lower. This results in a _____and a higher BOD removal efficiency in the system. **(lower, partial self digestion of the sludge)**
- 10: The efficiency of the system is comparable with_____, the operation is simpler, although less flexible. **(the conventional activated sludge system)**

Lec#117

- 11: High rate trickling filter similar to the low rate filters but receive a _____ **(higher BOD load per unit volume).**
- 12: _____is not digested in the filter. **(Sludge)**
- 13; Main objectives of recirculation _____. **(balancing the influent load);**



Read Only - You can't save changes to t...



availability is lower. This results in a _____ and a higher BOD removal efficiency in the system. **(lower, partial self digestion of the sludge)**

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Lec#117

11: High rate trickling filter similar to the low rate filters but receive a _____ **(higher BOD load per unit volume).**

12: _____ is not digested in the filter. **(Sludge)**

13; Main objectives of recirculation _____. **(balancing the influent load);**

57

14: Another way of improving the efficiency is by using _____ in series. This is called a _____. **(two filters, two-stage trickling filter system)**

15: Solid phase consists of a support medium and _____. **(biofilms)**

16: filters with stones are _____ meters in height, while filters with synthetic media can be _____ high. **(~3, more than 6 meters)**



Fariha shah

- This work is done by my Team , name

- Admin Arfa
- Shabana
- Sunbhal
- Maham
- Noor ul khuda
- Tassmia
- Aisha aish
- Eshal
- Mahwah
- Sanadeed
- Sabba urooj
- Shazine
- Zonish
- Sayeda zahra
- Zarnisha
- Andilla
- Faiza zahra
- Salma

- Dua kazmi
- Iqra zia

❖ I am highly thankful to my precious team for their Hard work.

❖ I will suggest all of my readers that prepare your handout thoroughly and deeply read them carefully

❖ Wish you best luck for Grand quiz. Remmber us in your precious prayers

❖ By Admin Tehreem& Arfa